

Unbounded Wind Fields for Stratospheric Balloon Simulation

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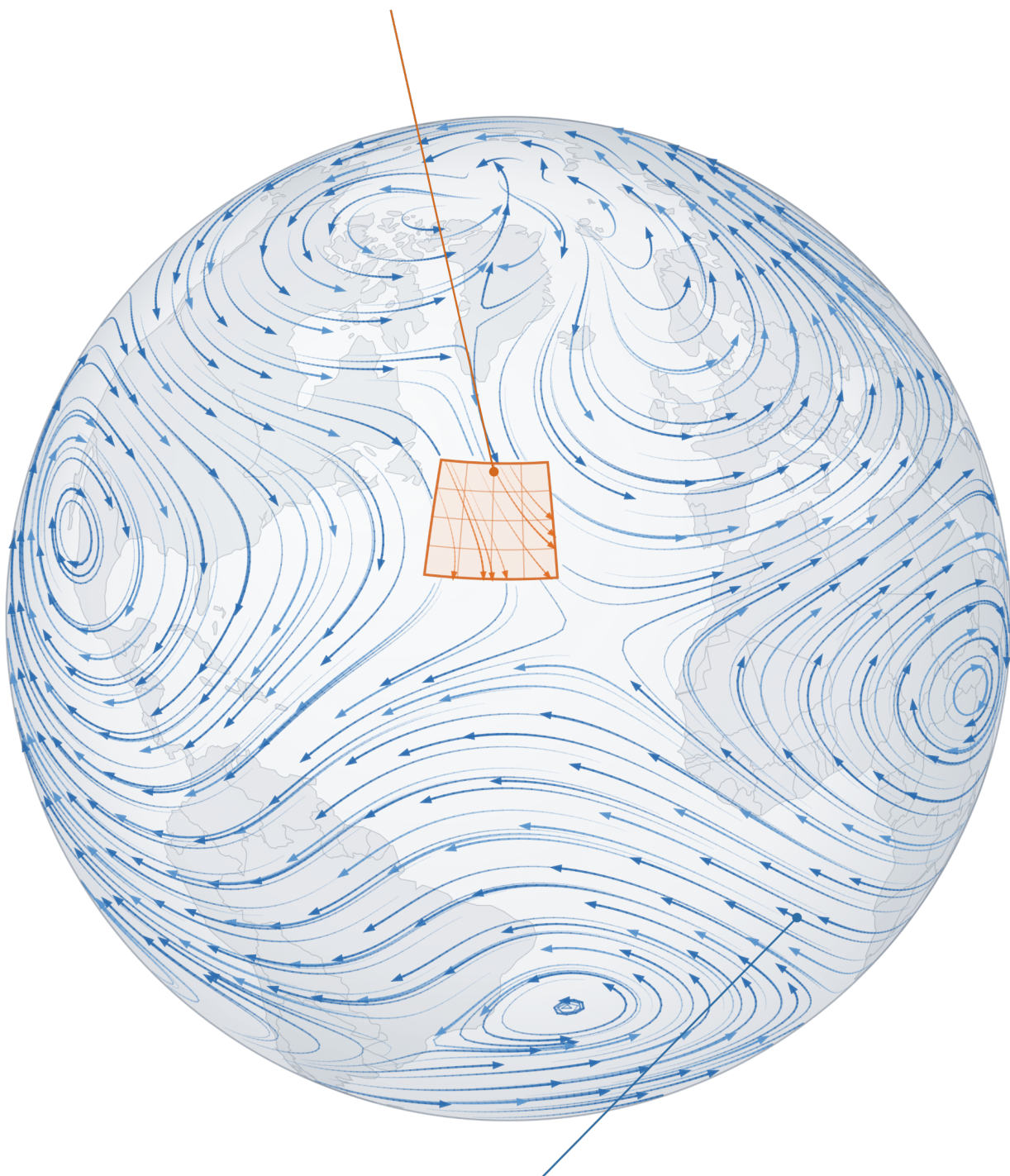
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github.com/sdean-group/balloon-env-dev

1 Why generate wind worlds?

Balloon simulators need winds that are both **realistic** and **effectively unbounded**.

Previous work



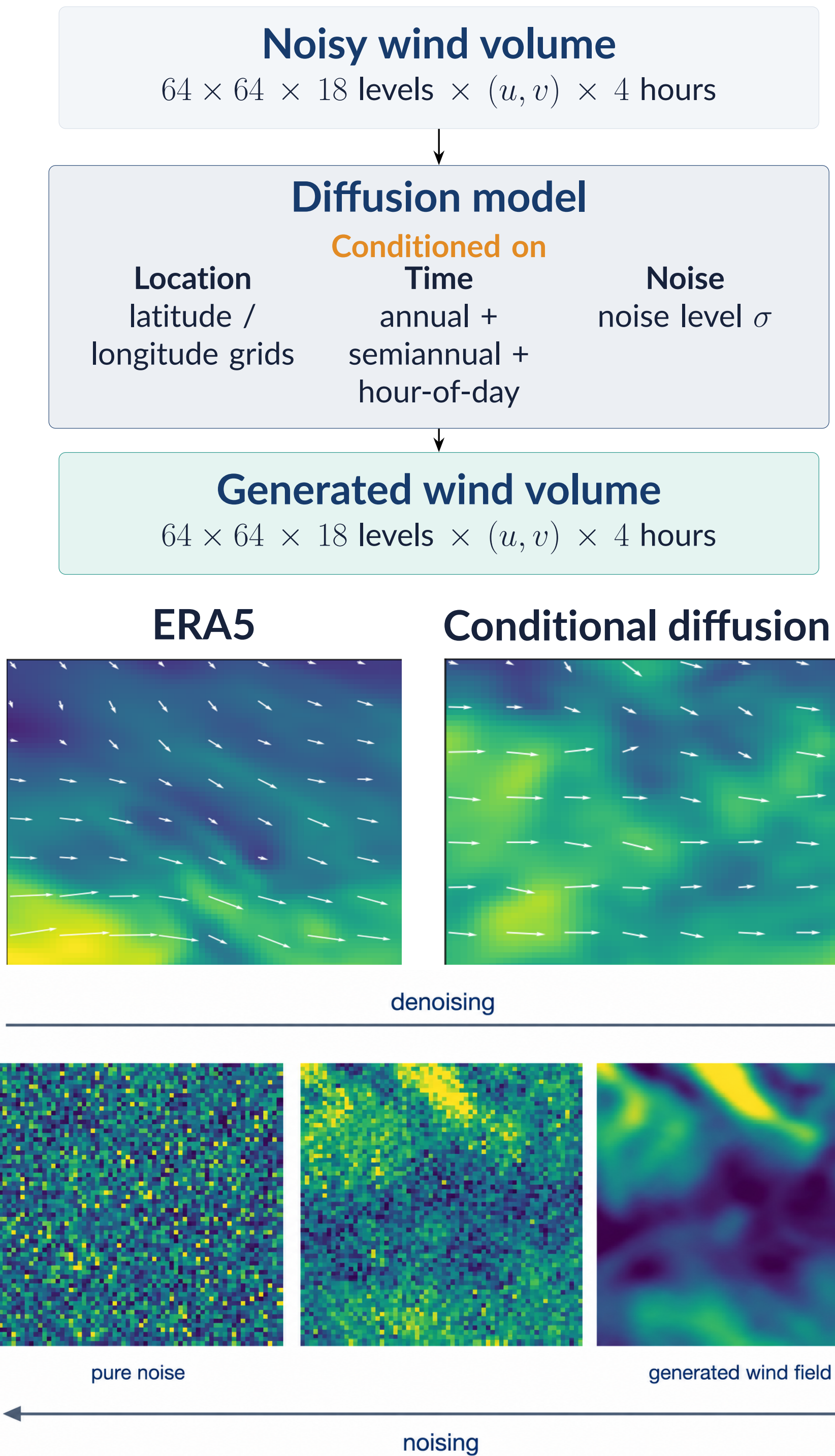
Our method

Lazily generate wind anywhere on the globe, across time

Our contributions

- A diffusion model for 64×64 ERA5 grids, 18 levels, and four hours.
- A lazy InfiniteDiffusion wrapper for reproducible, coordinate-indexed queries.
- A benchmark spanning spectra, marginals, extremes, vertical structure, and tiling.

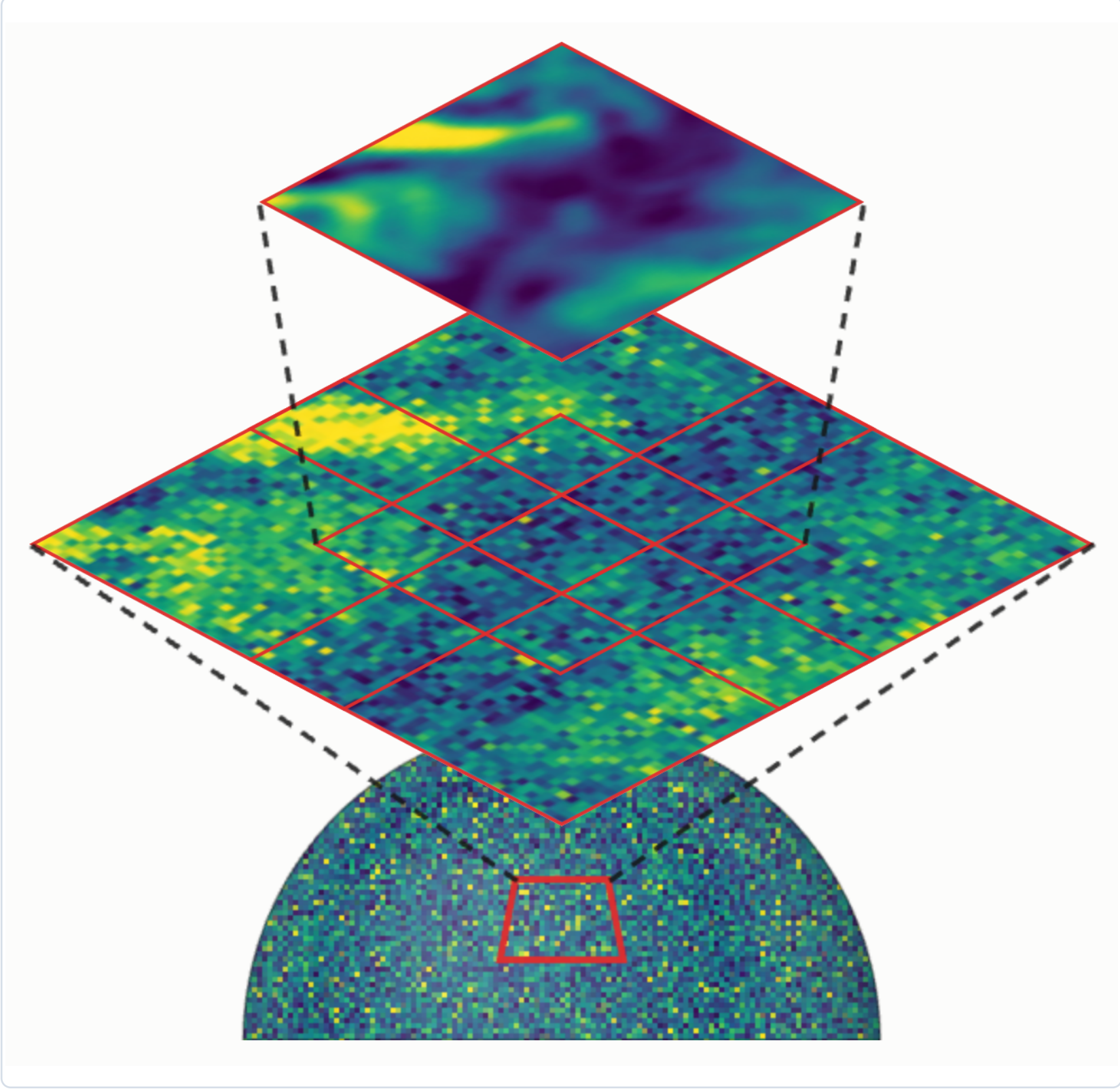
Base diffusion model



2 InfiniteDiffusion, one query at a time

A diffusion model only knows how to denoise one fixed-size window. **InfiniteDiffusion** reuses that local skill wherever wind is requested.

1. Cover the request with overlapping windows



2. Repeat the blend at every denoising step

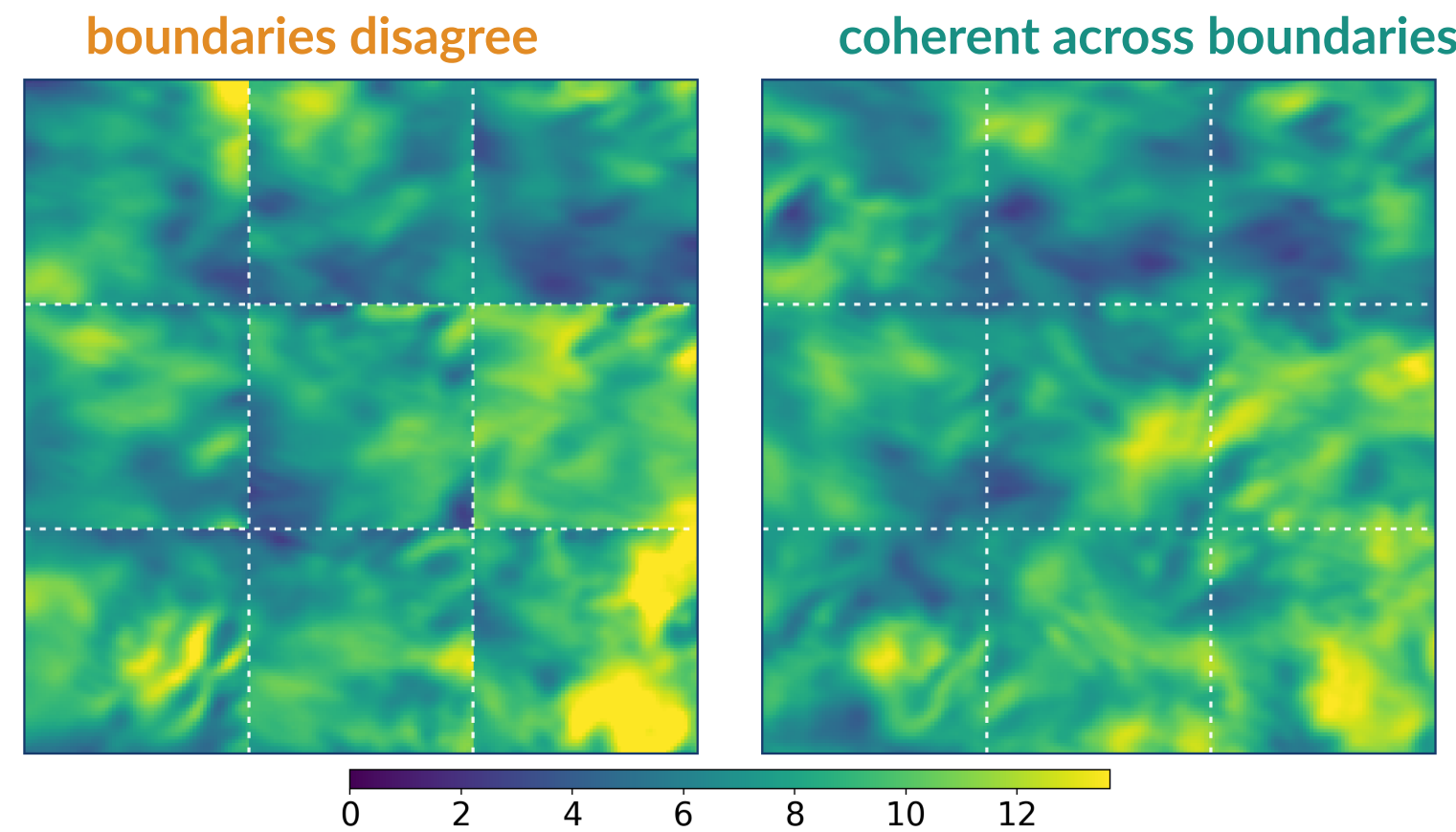
ALGORITHM 1 Query a wind region

Input: seed s , region R , denoising step t

- 1 if t is the pure-noise level:
- 2 **return** noise from s and global coordinates in R
- 3 weighted field $\leftarrow 0$; total weight $\leftarrow 0$
- 4 **for each** model window Q that overlaps R :
- 5 input $\leftarrow \text{Query}(s, Q, t + 1)$
- 6 prediction $\leftarrow \text{Denoise}(\text{input}, \text{position}(Q), \text{time})$
- 7 weight $\leftarrow$ mask that fades toward the edges of Q
- 8 weighted field $+= \text{weight} \times \text{prediction}$
- 9 total weight $+= \text{weight}$
- 10 result $\leftarrow \text{crop}(\text{weighted field} / \text{total weight}, R)$
- 11 cache result; **return** result

3 InfiniteDiffusion in action

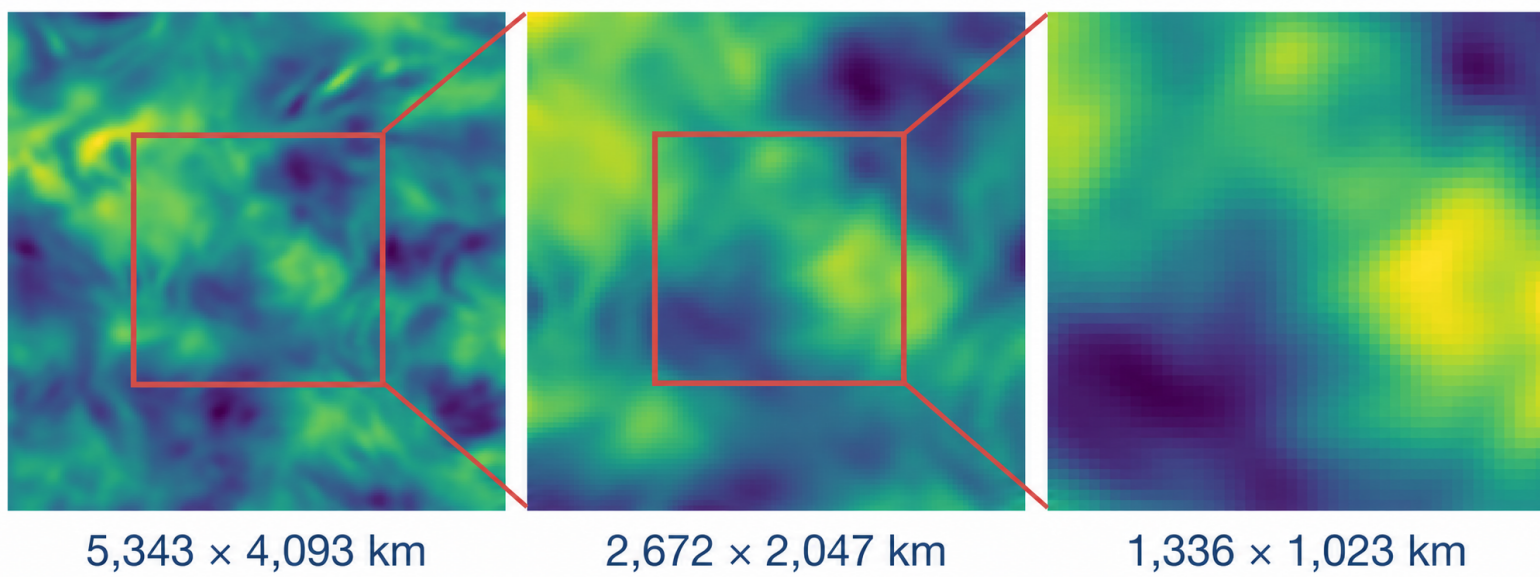
One bounded denoiser now behaves like a procedural wind field: **seamless across windows, repeatable, and queryable anywhere.**
Independent 64×64 samples InfiniteDiffusion



Guides mark 64×64 boundaries; trained-model tiling penalties are approximately zero across the reported benchmark.

$O(1)$ random access • deterministic • unbounded

Deterministic random access in practice



4 How we test realism

1. Distributions

Typical, conditional, and extreme winds.

2. Spatial physics

Energy by scale, divergence, and vorticity.

3. Vertical structure

Vertical shear across 18 pressure levels.

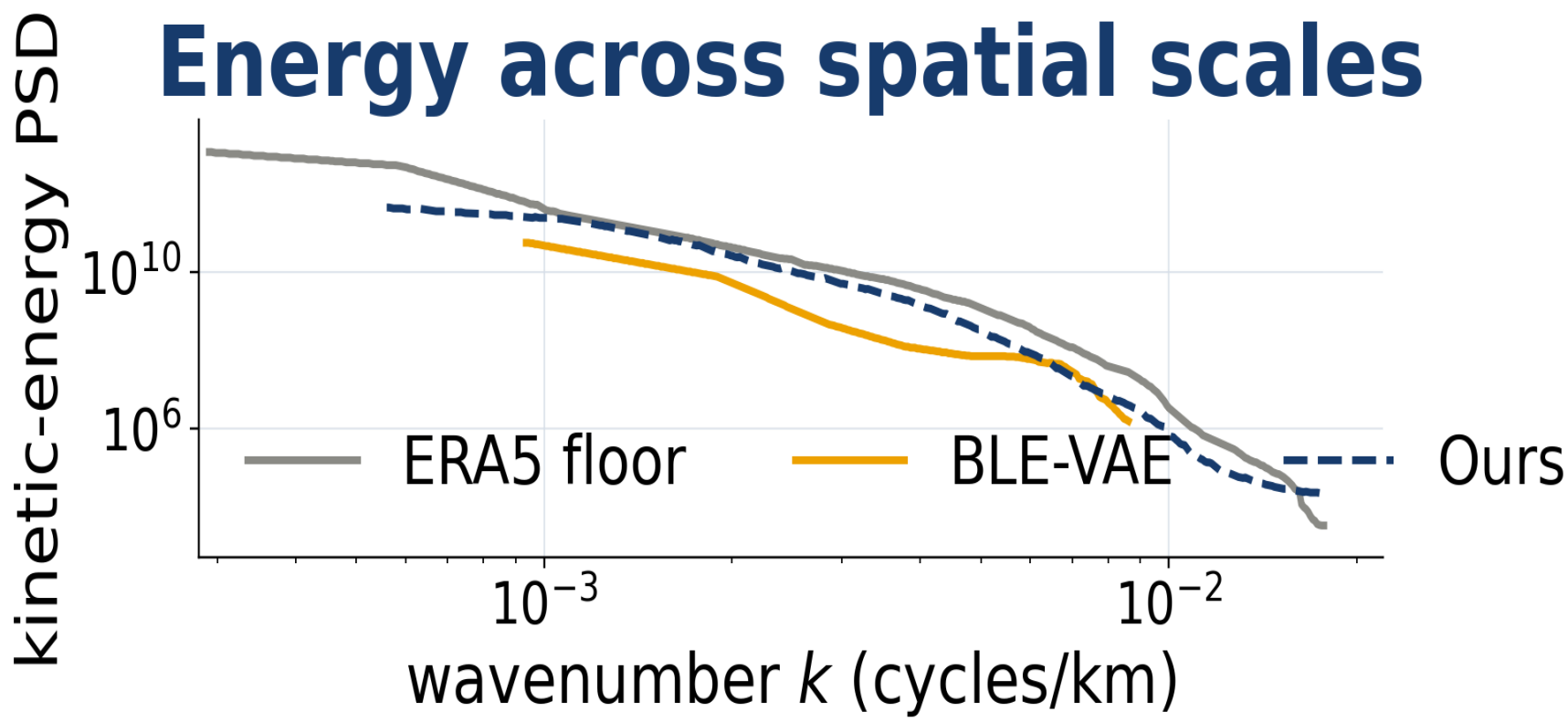
4. Temporal physics

Trajectory persistence and spread.

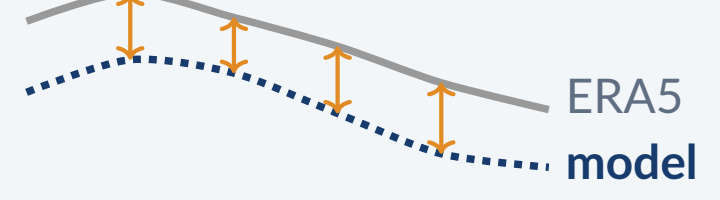
ERA5 floor: variation between held-out ERA5 subsets.

5 Results

TYPE	metric	floor	white noise	BLE-VAE	ours
DISTRIBUTIONS	$W_1((u, v) c)$ (m/s) ↓	2.38	3.13	N/A	3.52
	$W_1(V)$ (m/s) ↓	1.11	1.14	4.56	2.98
	tail error 0.1% ↓	2.92	2.59	18.94	13.54
SPATIAL PHYSICS	SR_E ↓	0.25	8.96	2.73	1.53
	SR_{div} ↓	0.25	8.98	5.23	1.56
	SR_{vort} ↓	0.25	8.99	2.16	1.53
	L_{eff} (km) ↓	56 [†]	3367	842	337
VERTICAL	W_1 shear u (m/s)/km ↓	0.20	18.44	3.62	1.00
	W_1 shear v (m/s)/km ↓	0.10	15.33	0.61	1.01
TEMPORAL	SR_{time} ↓	N/A	N/A	N/A	N/A
	trajectory spread	N/A	N/A	N/A	N/A

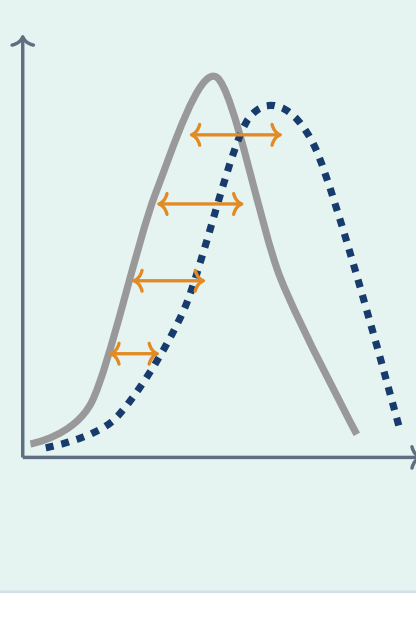


What does SR_E measure?



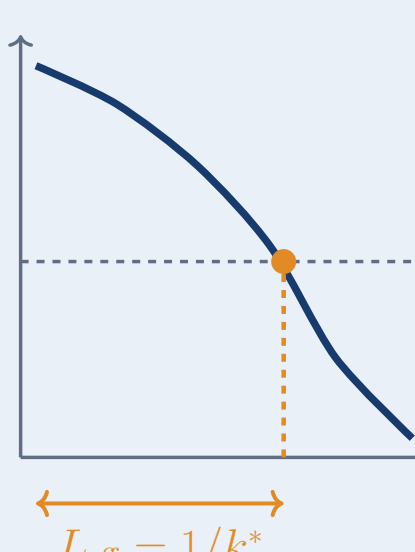
Mean log-scale gap between model and ERA5 energy spectra. **Lower is better.**

What does W_1 measure?



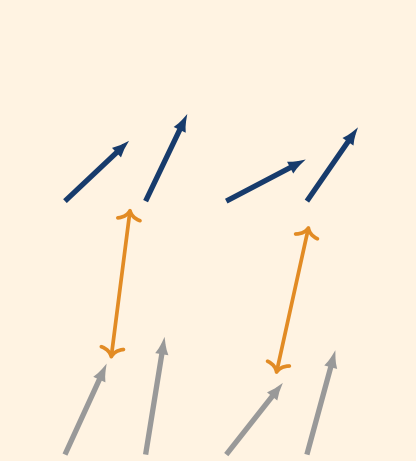
At fixed location, month, and hour c , compare ERA5 and model wind distributions. Average across conditions; **lower is better.**

What does L_{eff} measure?



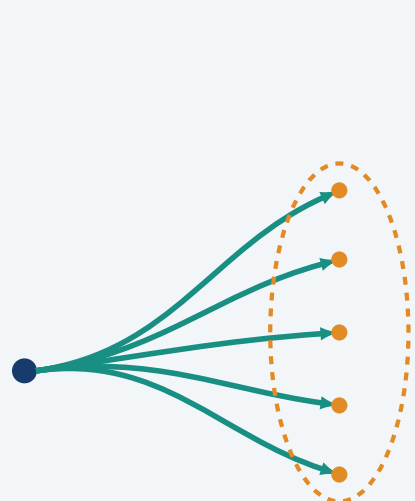
Smallest wavelength where the model retains at least half of ERA5 energy. **Lower is sharper.**
 $L_{eff} = 1/k^*$

What does W_1 shear measure?



W_1 between ERA5 and model distributions of $\Delta u / \Delta z$ and $\Delta v / \Delta z$. **Lower is better.**

What does trajectory spread measure?



Advert particles; compare their displacement and final spread with ERA5. **Closer is better.**

6 Conclusion

InfiniteDiffusion turns a fixed-size conditional denoiser into an effectively unbounded, repeatable wind field.

What worked

- Generate only requested regions.
- Revisit coordinates with identical winds.
- Retain location/time conditioning.

What remains

- Fine scales and extremes still differ from ERA5.
- Base-model realism is now the main bottleneck.

Next: coarse synoptic conditioning, stronger jet extremes, and arbitrary-length temporal generation.

Goslin et al., *InfiniteDiffusion* • Karras et al., EDM • Hersbach et al., ERA5